Combined LaTeX Transcriptions

Binder1.pdf Pages 1–20, followed by the first 9 standalone images

This document combines the earlier LaTeX transcriptions into one editable `.tex` file.

Standalone Images 1–9

Image 1

This is the 1^{st} generation only. We have two more copies of all these rows, identical in every way except for mass. 2^{nd} and 3^{rd} gen matter is unstable!

1^{st} **Gen**	2^{nd} **Gen**	3^{rd} **Gen**
Electron e	Muon μ	Tau τ
Electron Neutrino ν_e	Muon Neutrino ν_μ	Tau Neutrino ν_τ
Up Quark u	Charm Quark c	Top (Truth) Quark t
Down Quark d	Strange Quark s	Bottom (Beauty) Quark b

$$T_3 \equiv \tau_3/2$$

So, visually it's like:

$$Q = \frac{1}{2}(\tau_3 + Y)$$

rotate around z-axis   rotate around flagpole

Particle	**Electric Charge Q**	**Weak Isospin T_3**	**Hypercharge Y**
Electron e_R	-1	0	-2
Neutrino ν_e	0	$+1/2$	-1
Electron e_L	-1	$-1/2$	-1
Up Quark u_R	$+2/3$	0	$+4/3$
Down Quark d_R	$-1/3$	0	$-2/3$
Up Quark u_L	$+2/3$	$+1/2$	$+1/3$
Down Quark d_L	$-1/3$	$-1/2$	$+1/3$
$W^{\pm}$ Bosons	±1	±1	0
Z Boson	0	0	0
Photon γ	0	0	0
Gluon	0	0	0
Higgs H^+	$+1$	$+1/2$	$+1$
Higgs H^0	0	$-1/2$	$+1$

Leptons Quarks Gauge Bosons Higgs Field

$$SU(2)_L \text{ doublets: } \begin{pmatrix} \nu_e \\ e_L \end{pmatrix}, \quad \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}$$

$$\text{Gell-Mann–Nishijima relation:} \qquad Q = T_3 + \frac{Y}{2}$$

Image 2

$$\Psi = \begin{pmatrix} \psi_r \\ \psi_g \\ \psi_b \end{pmatrix}, \qquad \bar{\Psi} = \begin{pmatrix} \bar{\psi}_r & \bar{\psi}_g & \bar{\psi}_b \end{pmatrix}$$

Image 3

The color triplet field $\Psi \in \mathbb{C}^3 \otimes \mathbb{C}^4 \cong \mathbb{C}^{12}$ gives us a way to write $\mathcal{L}$ more concisely, but now that

we're here... what happens if we require $\mathcal{L}$ to remain unchanged under any transformation

$$\Psi \to \Psi' = U(x)\Psi \qquad \text{where } U(x) \in SU(3)?$$

Image 4

Free quark, non-interacting three-color Dirac lagrangian:

$$\begin{aligned}
&\mathcal{L} = \mathcal{L}_r + \mathcal{L}_g + \mathcal{L}_b \\
&\mathcal{L} = \bar{\psi}_r \left(i\gamma^\mu \partial_\mu - m \right) \psi_r + \bar{\psi}_g \left(i\gamma^\mu \partial_\mu - m \right) \psi_g + \bar{\psi}_b \left(i\gamma^\mu \partial_\mu - m \right) \psi_b
\end{aligned}$$

$$\mathcal{L} = \bar{\Psi} \left(i\gamma^\mu \partial_\mu - m \right) \Psi$$

Image 5

Free quark, non-interacting three-color Dirac lagrangian:

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_r + \mathcal{L}_g + \mathcal{L}_b \\
\mathcal{L} &= \bar{\psi}_r \left(i\gamma^\mu \partial_\mu - m \right) \psi_r \\
&\quad + \bar{\psi}_g \left(i\gamma^\mu \partial_\mu - m \right) \psi_g \\
&\quad + \bar{\psi}_b \left(i\gamma^\mu \partial_\mu - m \right) \psi_b
\end{aligned}$$

Image 6

$$\begin{aligned}
\psi_r &= u_s(\vec{p}) \exp[-i \left(Et - \vec{p} \cdot \vec{x} \right)] \,, \\
\psi_g &= u_s(\vec{p}) \exp[-i \left(Et - \vec{p} \cdot \vec{x} \right)] \,, \\
\psi_b &= u_s(\vec{p}) \exp[-i \left(Et - \vec{p} \cdot \vec{x} \right)] \,.
\end{aligned}$$

Image 7

Baryon	**Quark content**	**Charge**	**Mass** (GeV/c^2)	**Half Life (seconds)**	**Principal decays**	**Spin**
p	uud	$+1$	0.93828	at least $3 \cdot 10^{41}$	–	1/2
n	udd	0	0.93957	880	$pe\bar{\nu}_e$	1/2
Λ	uds	0	1.1156	$2.63 \cdot 10^{-10}$	$p\pi^-$, $n\pi^0$	1/2
Σ^+	uus	$+1$	1.1894	$0.80 \cdot 10^{-10}$	$p\pi^0$, $n\pi^+$	1/2
Σ^0	uds	0	1.1925	$6 \cdot 10^{-20}$	$\Lambda\gamma$	1/2
Σ^-	dds	-1	1.1973	$1.48 \cdot 10^{-10}$	$n\pi^-$	1/2
Ξ^0	uss	0	1.3149	$2.90 \cdot 10^{-10}$	$\Lambda\pi^0$	1/2
Ξ^-	dss	-1	1.3213	$1.64 \cdot 10^{-10}$	$\Lambda\pi^-$	1/2
Λ_c^+	udc	$+1$	2.281	$2 \cdot 10^{-13}$	$pK^-\pi^+$	1/2
Δ	uuu, uud, udd, ddd	$+2$, $+1$, 0, -1	1.232	$0.6 \cdot 10^{-23}$	$N\pi$	3/2
Σ^*	uus, uds, dds	$+1$, 0, -1	1.385	$2 \cdot 10^{-23}$	$\Lambda\pi$, $\Sigma\pi$	3/2
Ξ^*	uss, dss	0, -1	1.533	$7 \cdot 10^{-23}$	$\Xi\pi$	3/2
Ω^-	sss	-1	1.672	$0.82 \cdot 10^{-10}$	ΛK^-, $\Xi^0\pi^-$, $\Xi^-\pi^0$	3/2

Source: *Introduction to Elementary Particles*, David Griffiths, table at the beginning of the book.

Image 8

Vector Meson (Spin-1)	**Quark content**	**Charge**	**Mass** (GeV/c^2)	**Half Life (seconds)**	**Principal decays**
ρ	$u\bar{d}$, $d\bar{u}$, $(u\bar{u}-d\bar{d})/\sqrt{2}$	+1, −1, 0	0.770	$0.4 \cdot 10^{-23}$	$\pi\pi$
K^*	$u\bar{s}$, $s\bar{u}$, $d\bar{s}$, $s\bar{d}$	+1, −1, 0, 0	0.892	$1 \cdot 10^{-23}$	$K\pi$
ω	$(u\bar{u}+d\bar{d})/\sqrt{2}$	0	0.783	$7 \cdot 10^{-23}$	$\pi^+\pi^-\pi^0$, $\pi^0\gamma$
ϕ	$s\bar{s}$	0	1.020	$20 \cdot 10^{-23}$	K^+K^-, $K^0\bar{K}^0$
J/ψ	$c\bar{c}$	0	3.097	$1 \cdot 10^{-20}$	e^+e^-, $\mu^+\mu^-$, 5π, 7π
D^*	$c\bar{d}$, $d\bar{c}$, $c\bar{u}$, $u\bar{c}$	+1, −1, 0, 0	2.010	$> 1 \cdot 10^{-22}$	$D\pi$, $D\gamma$
Υ	$b\bar{b}$	0	9.460	$2 \cdot 10^{-20}$	$\tau^+\tau^-$, $\mu^+\mu^-$, e^+e^-

Image 9

Pseudoscalar Meson (Spin-0)	**Quark content**	**Charge**	**Mass** (GeV/c^2)	**Half Life (seconds)**	**Principal decays**
$\pi^\pm$	$u\bar{d}$, $d\bar{u}$	+1, −1	0.139569	$2.60 \cdot 10^{-8}$	$\mu\nu_\mu$
π^0	$(u\bar{u}-d\bar{d})/\sqrt{2}$	0	0.134964	$8.7 \cdot 10^{-17}$	$\gamma\gamma$
$K^\pm$	$u\bar{s}$, $s\bar{u}$	+1, −1	0.49367	$1.24 \cdot 10^{-8}$	$\mu\nu_\mu$, $\pi^\pm\pi^0$, $\pi^\pm\pi^\pm\pi^\mp$
K^0, $\bar{K}^0$	$d\bar{s}$, $s\bar{d}$	0, 0	0.49772	$K^0_S : 0.892 \cdot 10^{-10}$	$\pi^+\pi^-$, $\pi^0\pi^0$
K^0, $\bar{K}^0$	$d\bar{s}$, $s\bar{d}$	0, 0	0.49772	$K^0_L : 5.18 \cdot 10^{-8}$	$\pi e\nu_e$, $\pi\mu\nu_\mu$, $\pi\pi\pi$
η	$(u\bar{u}+d\bar{d}-2s\bar{s})/\sqrt{6}$	0	0.5488	$7 \cdot 10^{-19}$	$\gamma\gamma$, $\pi^0\pi^0\pi^0$, $\pi^+\pi^-\pi^0$
η'	$(u\bar{u}+d\bar{d}+s\bar{s})/\sqrt{3}$	0	0.9576	$3 \cdot 10^{-21}$	$\eta\pi\pi$, $\rho^0\gamma$
$D^\pm$	$c\bar{d}$, $d\bar{c}$	+1, −1	1.869	$9 \cdot 10^{-13}$	$K\pi\pi$
D^0, $\bar{D}^0$	$c\bar{u}$, $u\bar{c}$	0, 0	1.865	$4 \cdot 10^{-13}$	$K\pi\pi$
$D^\pm_s$	$c\bar{s}$, $s\bar{c}$	+1, −1	1.971	$3 \cdot 10^{-13}$	not established
$B^\pm$	$u\bar{b}$, $b\bar{u}$	+1, −1	5.271	$14 \cdot 10^{-13}$	$D+?$
B^0, $\bar{B}^0$	$d\bar{b}$, $b\bar{d}$	0, 0	5.275	$14 \cdot 10^{-13}$	$D+?$
η_c	$c\bar{c}$	0	2.981	$6 \cdot 10^{-23}$	$KK\pi$, $\eta\pi\pi$, $\eta'\pi\pi$

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Flow on a Rotating Planet – Horizontal Scaling

Euler Equation in Rotating Frame:

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho}$$

x-component of the Euler equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + 2\Omega w\cos\theta - 2\Omega v\sin\theta = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

Assume hydrostatic pressure scale:

$$\Delta P \sim \rho' gH$$

horizontal velocity scale U, timescale $T = L/U$,

$$w \sim \frac{UH}{L}$$

Scales of terms:

$$\frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + 2\Omega\frac{UH}{L} - 2\Omega U \sim \frac{\rho' gH}{\rho L}$$

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Flow on a Rotating Planet – Horizontal Scaling in x

x-component of the Euler equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + 2\Omega w\cos\theta - 2\Omega v\sin\theta = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

$$\frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + 2\Omega\frac{UH}{L} - 2\Omega U \sim \frac{\rho' gH}{\rho L}$$

$$1 + 1 + 1 + 1 + \frac{2\Omega L}{U}\left(\frac{H}{L} - 1\right) \sim \frac{\rho' gH}{\rho U^2}$$

Rossby Number:

$$Ro = \frac{U}{2\Omega L}$$

$\rightarrow$ ratio of inertial to rotation timescales

$Ro \ll 1$ for "slow" flow, "large" lengthscales and "rapid" rotation rates

$Ro \ll 1 \qquad \rightarrow \qquad$ Coriolis balances pressure gradient

$$2\Omega v\sin\theta = \frac{1}{\rho}\frac{\partial p}{\partial x}$$

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Flow on a Rotating Planet – Horizontal Scaling in y

y-component of the Euler equation:

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + 2\Omega u \sin\theta = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

Scales of terms:

$$\frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + 2\Omega U \sim \frac{\rho' g H}{\rho L}$$

$$1 + 1 + 1 + 1 + \frac{2\Omega L}{U} \sim \frac{\rho' g H}{\rho U^2}$$

Again, Rossby Number:

$$Ro = \frac{U}{2\Omega L}$$

$\rightarrow$ ratio of inertial to rotation timescales

$Ro \ll 1$ for "slow" flow, "large" length scales and "rapid" rotation rates

$Ro \ll 1 \quad \rightarrow \quad$ Coriolis balances pressure gradient

$$-2\Omega u \sin\theta = \frac{1}{\rho}\frac{\partial p}{\partial y}$$

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Navier–Stokes Momentum Equation in Rotating Frame

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho} + \nu\nabla^2\mathbf{u}$$

Euler Equation in Rotating Frame

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho}$$

$$2\Omega \times \mathbf{u} = (-fv,\ fu,\ 0)$$

$$f = 2\Omega \sin\theta$$

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Flow on a Rotating Planet – Vertical Scaling

Euler Equation in Rotating Frame:

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho}$$

Vertical component of the Euler equation:

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} - 2\Omega u \cos\theta = -g - \frac{1}{\rho}\frac{\partial p}{\partial z}$$

$$H \ll L$$

$$T = \frac{L}{U}$$

$$\frac{U}{L} \sim \frac{W}{H} \qquad \Rightarrow \qquad W \sim \frac{UH}{L}$$

Page 1

Planetary Rossby Waves

Begin with the assumptions for conserving potential vorticity: inviscid, thin, “rapid” rotation.

$$\frac{D}{Dt}\left(\frac{f+\zeta}{h}\right) = 0$$

The equations describing horizontal motions are:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{3}$$

Consider fluctuations with no mean flow (e.g. waves) and small velocities (small Ro), so neglect nonlinear advection terms.

Page 2

$$\frac{\partial u}{\partial t} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

Substitute for $f = f_0 + \beta y$,

$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

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$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

Replace pressure gradients with surface height anomalies η, that is,

$$\frac{1}{\rho}\frac{\partial p}{\partial x} = g\frac{\partial \eta}{\partial x} \qquad \text{(hydrostatic)}$$

so that

$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -g\frac{\partial \eta}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -g\frac{\partial \eta}{\partial y} \tag{2}$$

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$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -g\frac{\partial \eta}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -g\frac{\partial \eta}{\partial y} \tag{2}$$

Recall that the flow is predominantly geostrophic, so the time-dependent and beta terms are small compared to geostrophy.

As the geostrophic terms dominate, use these to express the velocities as

$$u \approx -\frac{g}{f_0}\frac{\partial \eta}{\partial y}, \qquad v \approx \frac{g}{f_0}\frac{\partial \eta}{\partial x}$$

and substitute back into (1,2) for the small terms:

$$-\frac{g}{f_0}\frac{\partial^2 \eta}{\partial y\,\partial t} - f_0 v - \beta y \frac{g}{f_0}\frac{\partial \eta}{\partial x} = -g\frac{\partial \eta}{\partial x} \tag{4}$$

$$\frac{g}{f_0}\frac{\partial^2 \eta}{\partial x\,\partial t} + f_0 u - \beta y \frac{g}{f_0}\frac{\partial \eta}{\partial y} = -g\frac{\partial \eta}{\partial y} \tag{5}$$

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Rearrange (4,5) for v and u:

$$v = \frac{g}{f_0}\frac{\partial \eta}{\partial x} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial y\,\partial t} - \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial x} \tag{6}$$

$$u = -\frac{g}{f_0}\frac{\partial \eta}{\partial y} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial x\,\partial t} + \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial y} \tag{7}$$

The blue terms are ageostrophic. These describe the behaviour of the wave perturbations superimposed on the geostrophic flow.

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Recall (3), the continuity equation,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{3}$$

This can be integrated over a layer depth H,

$$\int_0^H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) dz = H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + w(H) - w(0) = 0$$

$$\frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{8}$$

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Substitute u, v from (6,7) into (8):

$$v = \frac{g}{f_0}\frac{\partial \eta}{\partial x} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial y\,\partial t} - \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial x}, \tag{1}$$

$$u = -\frac{g}{f_0}\frac{\partial \eta}{\partial y} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial x\,\partial t} + \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial y} \tag{6,7}$$

$$\frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{8}$$

$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(\frac{\partial}{\partial x}\left[-f_0\frac{\partial \eta}{\partial y} - \frac{\partial^2 \eta}{\partial x\,\partial t} + \beta y\frac{\partial \eta}{\partial y}\right] + \frac{\partial}{\partial y}\left[f_0\frac{\partial \eta}{\partial x} - \frac{\partial^2 \eta}{\partial y\,\partial t} - \beta y\frac{\partial \eta}{\partial x}\right]\right) = 0$$

$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(-\frac{\partial^3 \eta}{\partial x^2\,\partial t} - \frac{\partial^3 \eta}{\partial y^2\,\partial t} - \beta\frac{\partial \eta}{\partial x}\right) = 0$$

$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(-\frac{\partial}{\partial t}\left[\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}\right] - \beta\frac{\partial \eta}{\partial x}\right) = 0$$

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$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(-\frac{\partial}{\partial t}\left[\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}\right] - \beta\frac{\partial \eta}{\partial x}\right) = 0$$

Substitute for

$$\nabla^2 \eta = \frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}$$

and define

$$R = \frac{\sqrt{gH}}{f_0}$$

as the Rossby radius:

$$\frac{\partial \eta}{\partial t} - R^2\frac{\partial}{\partial t}\nabla^2 \eta - R^2\beta\frac{\partial \eta}{\partial x} = 0 \tag{9}$$

This is the Rossby wave equation.

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$$\int_0^H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) dz = \frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0$$

$$\frac{1}{\rho}\frac{\partial p}{\partial x} = g\frac{\partial \eta}{\partial x} \qquad \text{(hydrostatic)}$$

$$R = \frac{\sqrt{gH}}{f_0} \qquad \text{as the Rossby radius}$$

$$\frac{\partial \eta}{\partial t} - R^2\frac{\partial}{\partial t}\nabla^2\eta - R^2\beta\frac{\partial \eta}{\partial x} = 0$$

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$$\frac{\partial \eta}{\partial t} - R^2\frac{\partial}{\partial t}\nabla^2\eta - R^2\beta\frac{\partial \eta}{\partial x} = 0 \tag{9}$$

This is the Rossby wave equation.

The coefficients of (9) are constant, and we have assumed the perturbation is a wave, so we prescribe the form

$$\eta = \eta_0\cos(kx - my - \omega t).$$

This allows us to rewrite (9) in terms of ω as a dispersion relation for Rossby waves:

$$\omega = -\beta R^2\frac{k}{1 + R^2(k^2 + m^2)}$$

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Dispersion relation for planetary Rossby waves

$$\left(\text{recall } R = \sqrt{gH}/f_0\right):$$

$$\omega = -\beta R^2\frac{k}{1 + R^2(k^2 + m^2)}$$

- For $\beta = 0 \to \omega = 0$: steady, geostrophic flow at constant f (no waves).
- We have assumed $Ro \ll 1$, so solution only valid for $\omega \ll f_0$.
- Consider $k, m \sim 1/L \to \omega \sim \beta R^2 L/(L^2 + R^2)$.
- So for short waves $L < R$: $\omega \sim \beta L$.
- For long waves $L > R$: $\omega \sim \beta R^2/L < \beta L$.
- For $L = R$: ω is maximum at $\omega_{\max} = \beta R/2$.

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Dispersion relation for planetary Rossby waves

$$\left(\text{recall } R = \sqrt{gH}/f_0\right):$$

$$\omega = -\beta R^2 \frac{k}{1 + R^2(k^2 + m^2)}$$

- ZONAL phase speed always negative $\rightarrow$ westward propagation

$$c_x = \frac{\omega}{k} = -\beta R^2 \frac{1}{1 + R^2(k^2 + m^2)}$$

- MERIDIONAL phase speed either sign

$$c_y = \frac{\omega}{m} = -\beta R^2 \frac{k/m}{1 + R^2(k^2 + m^2)}$$

- Very long waves ($1/k, 1/m, L \gg R$): propagate westward at

$$c = -\beta R^2,$$

  maximum speed allowed.

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$$\frac{D}{Dt}\left(\frac{f + \zeta}{h}\right) = 0$$

North, high f South, low f

Contours of f

Greater f, reduced ζ Smaller f, increased ζ

Greater f, reduced ζ

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Taylor–Proudman Theorem

For steady, $Ro \ll 1$ and uniform density ρ.

Geostrophic vector equation:

$$2\Omega \times \mathbf{u} = -\frac{\nabla p}{\rho}$$

Take the curl of this equation:

$$\nabla \times (2\Omega \times \mathbf{u}) = -\nabla \times \left(\frac{\nabla p}{\rho}\right)$$

RHS: curl of a gradient (∇p) is zero, ρ is uniform.
So:

$$\nabla \times (\Omega \times \mathbf{u}) = 0$$

Expands to:

$$(\Omega \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\Omega + \mathbf{u}(\nabla \cdot \Omega) - \Omega(\nabla \cdot \mathbf{u}) = 0$$

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Taylor–Proudman Theorem

$$(\Omega \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\Omega + \mathbf{u}(\nabla \cdot \Omega) - \Omega(\nabla \cdot \mathbf{u}) = 0$$

Ω is uniform everywhere, and continuity is unaltered by rotation, so:

$$(\Omega \cdot \nabla)\mathbf{u} = 0$$

Choose coordinates such that Ω is in the z-direction,

$$\Omega\frac{\partial \mathbf{u}}{\partial z} = 0 \qquad \rightarrow \qquad \frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = \frac{\partial w}{\partial z} = 0$$

And for a domain with a solid (bottom) boundary where $w = 0$,

$$\frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = 0, \qquad w = 0$$

Horizontal velocities constant with depth, moves as “Taylor columns”.